MATHS I · WEEK 3

Maths I week 3 formulas: Quadratic functions and equations

Quadratic functions & parabolas · Quadratic equations & roots. Save this PDF for revision.

QUADRATIC FUNCTIONS & PARABOLAS · IN EVERY PAPER

Quadratic functions & parabolas

Standard form

$$f(x) = ax^2 + bx + c,\ a \neq 0$$

Vertex x

$$h = -b/2a$$

Vertex y

$$k = f(h) = -D/4a = c - b^2/4a$$

Axis of symmetry

$$x = -b/2a$$

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Quadratic functions & parabolas

Vertex form

$$f(x) = a(x - h)^2 + k$$

Opens

$a > 0$: up, minimum $= k$ · $a < 0$: down, maximum $= k$

Range

$a > 0$: $[k, \infty)$ · $a < 0$: $(-\infty, k]$

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Quadratic functions & parabolas

y-intercept

$(0, c)$

Slope at x (tangent)

$2ax + b$

Zero at the vertex.

Slope of secant $x_1 \to x_2$

$$(f(x_2) - f(x_1))/(x_2 - x_1) = a(x_1 + x_2) + b$$

Increasing / decreasing

$a > 0$: decreasing on $(-\infty, h)$, increasing on (h, ∞)

Reversed when $a < 0$.

Common traps

⚠ The max/min value is k (the y-coordinate). The input that gives it is h. Read which one is asked.

⚠ Vertex x is −b/(2a): keep both the minus sign and the 2.

⚠ $D < 0$ with $a > 0$ means the parabola is entirely above the x-axis, so $f(x) > 0$ for every x.

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Quadratic equations & roots

Roots

$$x = (-b \pm \sqrt{D})/2a,\ D = b^2 - 4ac$$

Nature of roots

D > 0: two distinct real · D = 0: equal, x = –b/2a · D < 0: no real roots

Rational roots

D is a perfect square (a, b, c rational)

Sum and product

$$\alpha + \beta = -b/a,\ \alpha\beta = c/a$$

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Quadratic equations & roots

Equation from roots

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

Sum of squares

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

Difference

$$|\alpha - \beta| = \sqrt{D}/|a|$$

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Quadratic equations & roots

Reciprocals

$$1/\alpha + 1/\beta = (\alpha + \beta)/\alpha\beta$$

Cubes

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

Signs of roots

$c/a < 0 \Rightarrow$ opposite signs · both positive $\Leftrightarrow -b/a > 0$ and $c/a > 0$

For real roots ($D \geq 0$).

Completing the square

$$ax^2 + bx + c = a(x + b/2a)^2 - D/4a$$

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Common traps

⚠ Tip: Roots given and the equation asked? Don't expand anything. Check each option's –b/a and c/a against the sum and product.

⚠ Wrong constant ⇒ the sum is still right. Wrong x-coefficient ⇒ the product is still right.

⚠ x-intercepts of the graph are the real roots. D < 0 ⇒ the graph never meets the x-axis.

Notes for every week of Maths I, free.

Full notes, the formula sheet and common traps, with worked PYQs.

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